Sahil Pathak

AI/ML and full-stack engineer

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SUMMARY

Final-year computer science undergraduate specialising in AI and data science, with deep learning research and production full-stack work. Co-developed a PyTorch multimodal model that reached 91.24% accuracy in depression detection, and cut a production MERN app's load time by 50 to 80%. Core stack: Python, JavaScript, TypeScript, React, Next.js, Node.js, TensorFlow, Hugging Face.

EXPERIENCE

Research intern, Vellore Institute of Technology (VIT) May 2026 to Jul 2026

- Analysed 20+ research papers (2020 to 2026) and led end-to-end assembly of a model fusing text, EEG, wearable and audio/video data.
- Co-developed the PPEMDD framework in PyTorch, pairing a gated fusion network (DGN) with multi-head cross-attention (MHCA) to perform three tasks in one model: depression classification, PHQ-9 severity regression and DSM-5 symptom recognition.
- Achieved 91.24% accuracy, 79.23% recall and a 0.7809 F1-score on a 2,250-sample held-out test set (Reddit and WU3D corpora), using a leakage-aware protocol with no test-set threshold tuning.
- Engineered availability-aware fusion that handles missing modalities.

Software developer, AMG Turnkey Projects LLP Jan 2025 to Jun 2025

- Owned development of the live production website (React, Vite, Tailwind CSS, Node.js, Express, MongoDB), built to make the company's construction services findable across three states.
- Worked with the operations team for six months to deliver civil accounting modules and REST APIs for project workflows.
- Diagnosed performance bottlenecks and optimised the application, cutting load time by 50 to 80% and improving reliability.

RESEARCH

PPEMDD, an IEEE-format research paper (2026) proposing an adaptive multimodal deep learning framework for depression detection with gated cross-modal attention fusion.

PROJECTS

PatchProof, benchmark for overfitting in automated program repair Python, AST mutation, sandboxed execution

- Built a reproducible benchmark that injects 139 bugs into Python functions by AST mutation and scores every repair on tests the repair tool never saw: with one test visible, 41% of the passing fixes from a search-based repairer are wrong. Code at github.com/sahil454521/Project.

Sarkari Saathi, voice AI for government schemes TypeScript, React, Express, Gnani Prisma and Timbre, LLM

- Built a voice-first app, for Gnani AI's Great Indian AI Internship Challenge, that finds the government schemes a person may qualify for from a conversation in Marathi, Hindi or English. Live at sarkari-saathi-kjgb.onrender.com.
- The LLM only extracts facts as zod-validated JSON; TypeScript rules checked against official pages decide eligibility across 13 schemes, so it cannot invent one. 4 to 6 questions, about 3 s per spoken turn, 76 tests.

AI code editor (NeuraCraft) Next.js, FastAPI, scikit-learn, Convex, Clerk

- Built a full-stack AI code editor (Monaco) with real-time inline suggestions for four Python ML libraries. Live at ai-compiler-eta.vercel.app.
- Trained a scikit-learn model to predict the ML framework in use, served through FastAPI on Render with a Next.js client on Vercel.
- Implemented Clerk authentication and Convex real-time storage; curated a growing coding-error dataset to retrain the model.

Desi Totes e-commerce platform (client project) React, Node.js, Express, MongoDB, Razorpay

- Launched a live store for a client with Razorpay payments in two currencies (INR and USD) and persistent carts. Live at desitotes.com.

EDUCATION

B.Tech, Computer Science and Engineering, DY Patil International University, Pune Aug 2023 to Aug 2027

B.Sc. (Hons.), Data Science and Artificial Intelligence, IIT Guwahati Aug 2023 to Aug 2027

Coursework: data structures and algorithms, OOP, DBMS, operating systems, computer networks, machine learning.

SKILLS

Languages	Python, JavaScript, TypeScript, SQL, C++, C, HTML, CSS
AI and ML	Machine learning, deep learning, NLP, multimodal learning, generative AI, diffusion models, model evaluation
ML libraries	PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, Diffusers, Pandas, NumPy
Web	React, Next.js, Node.js, Express, FastAPI, Flask, Redux, Tailwind CSS, Vite, REST APIs
Data and tools	MongoDB, Firebase Firestore, Convex, Git, GitHub, Postman, VS Code, Vercel, Render, CI/CD

ACHIEVEMENTS

- Smart India Hackathon: top 15 teams in the college round, in a six-member team.
- FOSS Hackathon 2025: collaborated on an open-source AI coding assistant that improved automated code correction.
- SharkIndia (Avenyou Club): won a prize for an AI-based solution.
- Generative AI certification.